

Stakeholder Engagement

Primary Stakeholders

- Smallholder farmers in Sub-Saharan Africa seeking agricultural advice
- Agricultural extension workers who provide advice using extension materials
- Agricultural researchers and document authors whose materials form the corpus

How Stakeholder Needs Shaped the Project

The problem statement was motivated by a documented mismatch: farmers describe problems in plain language ("why are my crop leaves turning yellow?") while extension documents use technical language ("nitrogen deficiency causing chlorosis in maize"). This vocabulary gap not a lack of relevant information is the core problem the project targets, which is why the team tested lexical (TF-IDF, BM25) and semantic (dense embedding) retrieval side by side rather than assuming one approach was sufficient.

Engagement to Date

This phase used the dataset and problem framing provided through the Kaggle competition; the team did not conduct direct interviews with farmers or extension workers during the development window.

Planned Future Engagement

Before any real-world deployment, the team recommends:

- Direct review of retrieved-document quality by agricultural extension experts
- Usability feedback from farmers/extension workers on whether retrieved documents actually answer their real questions, not just keyword-match them
- Region- and crop-specific evaluation to check the 695-document corpus's coverage gaps identified in the Data Card